

Reviewer Report and Revision Plan

An Adaptive Temporal Personal Knowledge Graph with Memory Consolidation and Contradiction-Aware Retrieval-Augmented Generation

Manuscript: draft 1, 5 pages, 25 references. Authors: Dr. P S Anu Rakhi, Maria Immanuel L, Vigneshwaran S (Vel Tech Rangarajan Dr. Sagunthala R&D; Institute of Science and Technology).

Review scope: research problem and framing, novelty positioning, methodology, experimental design, statistical validity, claim-evidence alignment, reference verification against the published record, and IEEE presentation.

How to read this document

Stage 1 is the critical review. Stage 2 is the prioritised checklist. The corrected paper itself is a separate artefact (IEEE_Conference_Paper_SecondBrain_v3.pdf) so it can be submitted directly. Stage 4 is the experimental plan that removes each [REQUIRES EXPERIMENT] marker. Stage 5 is the defensible novelty statement. Appendix A is the reference audit, which contains the single most urgent correction in this review.

Stage 1 - Critical Review

1.1 What the draft does well

The draft has a real argument, and it is the right one for this system. It does not present the implementation as the contribution; it identifies a failure mode (retrieval treats outdated and contradictory personal memories as equally current evidence), shows why existing metrics cannot see it (faithfulness is preserved because nothing is fabricated), and reports a metric that can (stale@1). That framing is stronger than most student drafts, which present a system tour and append a screenshot.

Three further strengths are worth protecting through revision. First, the architecture is described as *additive* and every component is switchable, which is exactly what makes a controlled comparison possible; the draft states this and uses it. Second, the limitations section is already honest: it names the synthetic corpus, the rule-based extraction, the missing end-to-end generation evaluation and the absent ablation. Third, the negative result is reported rather than hidden - Hit@5 saturates at 1.000 for all four systems, and the draft says so instead of leading with a flattering metric.

1.2 Critical weaknesses

The weaknesses below are ordered by how much they threaten acceptance. W1 to W3 are blocking; W4 to W7 materially weaken the paper; W8 onwards are presentational but still worth fixing before submission.

ID	Weakness	Why it matters
W 1	Two reported figures are not reproducible from the code. The draft states "Ingestion produced 4 detected conflicts" and gives baseline stale@1 = 0.091. Re-running the described system produces 2 conflicts and stale@1 = 0.182 (Table 1.1 below).	This is the one finding that must be fixed before anything else. A reviewer who re-runs the artefact and gets different numbers will distrust every other number in the paper.
W 2	Several references do not match the published record: one has the wrong venue, year and truncated author names; one has a mangled surname initial; several carry future years; at least one could not be located at all. See Appendix A.	Incorrect citations are unambiguous, cheap to check, and read as carelessness or as padding. Some of the uncited references in the current list are also irrelevant to the argument.
W 3	The contribution is presented as seven co-equal components with no statement of which is the paper's actual claim, and no table separating adopted techniques from proposed ones.	As written, a reviewer can reasonably read the paper as claiming novelty for RAG, hybrid retrieval, knowledge graphs, consolidation and unlearning individually - none of which is novel. The draft is vulnerable on exactly the point where it is actually defensible.
W 4	No research questions and no hypotheses are stated. The evaluation measures things, but nothing is being tested.	Without stated hypotheses, the results cannot be said to confirm or refute anything, and the per-category table has no stated prediction to match.
W 5	The evaluation is retrieval-only, and the paper's central claim is about answers.	The draft acknowledges this in one limitation sentence, but the gap is large: a reviewer will ask how a 5.9% MRR improvement changes what the user reads. This needs a designed experiment, not only an admission.
W 6	No ablation. Five post-retrieval mechanisms are bundled and compared against three baselines; which mechanism produces the effect is unknown.	The per-category table is suggestive but cannot substitute for a component-wise design. Reviewers of systems papers treat the ablation as mandatory.
W 7	Statistical treatment is absent. Rates such as stale@1 and duplicate rate are reported as if exact, from 11 questions and 45 returned results per system.	Some of the reported differences are a single question changing rank. The paper should report counts, and once the benchmark grows, confidence intervals and a paired test.
W 8	The phrase "machine unlearning" is used for a retrieval-store operation, and "hallucination-free" is used as a description of the system.	Unlearning has a specific meaning in the literature (removing the influence of training data from parameters). Equating the two invites a rejection on terminology. "Hallucination-free" is an absolute claim no retrieval-layer result can support.

ID	Weakness	Why it matters
W 9	Structure and numbering are inconsistent: Section V is interleaved out of order in the typeset output (V-B and V-C appear after VI-A), and Table III is followed by a caption that does not match its content.	Typesetting defects of this kind are read as evidence that the paper was not proofread, which undermines confidence in the measurements.
W 10	Related work is organised by topic but contains no comparison table and no explicit gap statement, and it omits temporal knowledge representation and memory-consolidation literature entirely.	“Existing work is orthogonal” is asserted rather than shown. A gap table forces the claim to be explicit and checkable.

Table 1.1 - Reproducibility check. Each draft figure was re-measured by running the code, with the pre-fix research modules restored (commit 38d9b20) as well as with the current ones.

Quantity in draft	Draft value	Measured (pre-fix)	Measured (current)	Verdict
Conflicts detected at ingestion	4	2	2	Not reproducible
Hybrid stale@1	0.091	0.182	0.182	Not reproducible
Graph stale@1	0.091	0.182	0.182	Not reproducible
Vanilla stale@1	0.364	0.364	0.364	Reproduces
MRR (vanilla / hybrid / adaptive)	0.833 / 0.944 / 1.000	0.833 / 0.944 / 1.000	0.833 / 0.944 / 1.000	Reproduces
Leak and duplicate rates	0.022 → 0.000	same	same	Reproduces
Adaptive latency	80 ms	76 ms	76 ms	Wall-clock; varies per run and per machine

The two irreproducible values share a cause. Both came from a run whose database held state beyond the benchmark corpus: conflict rows and consolidation state belonging to other memories changed the counts, and retrieval was not scoped to the corpus, so an unrelated memory could take a top-5 slot and move stale@1. The corrected pipeline evaluates the corpus against itself and pins the dense backend, which is why the current numbers are stable across runs. The paper must therefore report the re-measured values, and should state the configuration explicitly, because the effect being measured depends on it.

1.3 Unsupported or overreaching statements in the draft

Location	Statement	Problem	Replacement
Abstract	“whereas every baseline continues to return explicitly unlearned content”	True only in the model-free configuration. With a vector store the embedding is deleted and no mode can return the content. As written it reads as a general property.	State the configuration inside the claim.
Abstract / Intro	Counts “seven components” then “three claims” without connecting them	A reviewer cannot tell what is claimed to be new.	Add the adopted-vs-proposed table (now Table III) and state the contribution as coordinated integration plus the specific mechanisms.
Section I	“the system is hallucination-free and still incorrect”	Absolute claim about generation, and no generation evaluation was run.	“every retrieved passage is faithful to its source, so a faithfulness metric cannot flag the error”.
Section II-D	“Machine Unlearning” as the heading for retrieval-store removal	Conflates two different problems [Bourtole et al. define unlearning over model parameters].	Rename to retrieval-level forgetting, and state the distinction explicitly.

Location	Statement	Problem	Replacement
Section IV-D	“the 0.90 threshold commonly recommended in practice missed every reworded duplicate”	No source is given for the recommendation, and “in practice” is unattributed.	Either cite a source or rewrite as the observation it is: our corpus's reworded duplicates scored ≈ 0.82 , so a 0.90 cutoff would merge nothing.
Section VI	“The proposed system therefore never leads with an outdated fact on this corpus”	Defensible as written (it says <i>on this corpus</i>), but the following sentence “roughly one query in eleven” understates the measured value.	Use the re-measured rate and its count: 2 of 11 questions.
Section VII	“the paper's claims are each mapped to at least one executable check”	True for the implementation, but the 68 assertions do not test any of the hypotheses; they test component behaviour.	Say that the suite tests component behaviour; the benchmark tests the hypotheses.

1.4 Novelty risks

There are three places where a reviewer could reject the paper for novelty, and one place where the paper is too modest.

- **Risk 1 - “this is engineering, not research.”** The draft's own first contribution is “a seven-component personal memory architecture”. Assembly of known parts is not a research contribution. The fix is not to inflate the components but to move the claim: the contribution is the *evaluation of a neglected failure mode* together with a benchmark that exposes it and an architecture whose switches let that failure be isolated. The paper's own finding - that the standard metric is saturated and therefore uninformative - is the interesting part and should be foregrounded.
- **Risk 2 - overlapping with agent-memory work.** Lifelong-learning and agent-memory roadmaps already treat evolving knowledge as a first-class concern. The paper must cite that literature (it currently does not) and state precisely what is added: an operational definition of *current* truth over the store, enforced at retrieval time, with the derived-surface problem made explicit.
- **Risk 3 - “temporal knowledge graphs are established.”** Correct, and the draft should say so. What is not established is using interval validity to decide between competing personal assertions at retrieval time, and measuring the effect with `stale@1`.
- **Being too modest.** The leakage result - that deletion at one surface is insufficient, demonstrated with the baselines still returning removed content - is a genuinely useful negative finding for personal AI, and the draft buries it in a limitations bullet. It belongs in the contributions.

1.5 Methodology risks

- **Corpus size.** 15 memories and 11 questions give per-category figures based on one to five questions. The draft's per-category table (its Table IV) therefore reports MRR deltas that a single question's rank change can produce. The paper should say this explicitly and treat the per-category deltas as directional.
- **Threshold and weight selection.** The consolidation threshold (0.78) was calibrated on the same corpus used for evaluation. The draft argues the threshold was chosen from the measured distribution rather than by inspecting the test set, which is the right instinct, but on a 15-memory corpus the two are hard to separate. State it as a limitation and repeat the calibration on a larger corpus.
- **Single-run reporting.** The draft reports one execution. Reproducibility is claimed on the basis that two consecutive runs agree, which is necessary but not sufficient: determinism of one configuration is not robustness across corpora.
- **Backend dependence.** The direction of the leakage result depends on whether a vector store is present, because a deleted embedding cannot be returned. This is not a flaw, but it must be stated in the claim itself rather than in a footnote.

1.6 IEEE presentation issues

- Section order in the typeset output is broken: Section V's subsections (Corpus, Systems Compared, Metrics, Reproducibility) appear after Section VI's first subsection. Section VI-A is then empty.
- Table III's caption is followed immediately by unrelated body text; the table's content is missing from the flow.
- Tables II to V are referenced in the text before the tables that define their columns appear.
- Equation (1) is reported as " $M = w_1R + w_2F + w_3T + w_4G + w_5U + w_6P - w_7D - w_8C$ (1)" with no definition list in the manuscript; the eight weights are never given, so the model is not reproducible from the paper.
- The T (recency) term is described as "exponential decay" with a 45-day half-life, but the expression is only stated as $T = 2^{-\Delta t/45}$ in the typeset text, with unclear minus-sign handling.
- Reference list: mixed formatting (some entries omit pages, one contains a mistyped ampersand as "S&P;"), future years (2026) for entries that do not yet exist, and, verified by scanning the body text for citation markers, ten of the 25 references are never cited at all: only [1]-[11], [16], [17], [20] and [23] appear in the argument.
- Terminology is inconsistent: "Second Brain", "SecondBrain" and "Personal Brain-Bench" versus "PersonalBrain-Bench" appear in the abstract alone.
- The Acknowledgment thanks a co-author (the supervisor is listed as an author) - remove.

Stage 2 - Required Changes (prioritised)

Red - must fix before submission

#	Change	Where
R1	Replace every irreproducible figure with the re-measured value: conflicts 2 (not 4); hybrid/graph stale@1 0.182 (not 0.091); latency from the same run as the rest of the table.	Abstract, V-A, VI, Tables II-V
R2	Correct or remove the references that do not match the published record, and renumber in order of first citation. Appendix A lists each one.	References
R3	Add the research question (one main, four specific) and hypotheses H1-H4, each mapped to a reported measurement.	Introduction
R4	Add the adopted-versus-proposed table so no established technique is presented as new, and rewrite the contribution list around coordinated integration plus the specific mechanisms.	Introduction, Section III
R5	State the evaluation configuration inside every claim that depends on it (dense backend, corpus scope, k), and state that the leakage asymmetry holds in that configuration.	Abstract, V-D, VI
R6	Replace “machine unlearning” with retrieval-level forgetting throughout, with one explicit sentence distinguishing them; delete “hallucination-free”.	Abstract, I, II, IV-F
R7	Fix the section ordering, the missing Table III content, and the empty VI-A in the typeset output.	Whole paper
R8	Give the eight weights, the normalisation ranges and the half-life formula in the manuscript so the scoring model is reproducible from the paper.	Section IV-A

Orange - strongly recommended

#	Change	Where
O1	Add the component-wise ablation design (cumulative, one mechanism per row) and mark it [REQUIRES EXPERIMENT].	New section
O2	Add the failure-case analysis: stating where the rules fail is what makes the rest of the claims credible.	New section
O3	Add conflict-detection precision/recall as a marked missing experiment; do not claim accuracy that has not been measured.	Section IV-C
O4	Add the end-to-end generation evaluation design, and say plainly that retrieval metrics are proxies.	Planned evaluation
O5	Add the statistical protocol (counts now; bootstrap CIs and a paired test once the benchmark grows).	Planned evaluation
O6	Expand Related Work to nine areas including temporal knowledge representation, consolidation and agent memory, and add the gap table.	Section II
O7	Report rates as counts alongside proportions, and mark where a difference is one question changing rank.	Section VI

Yellow - optional / future work

#	Change	Where
Y1	Expand the benchmark beyond 15 memories / 11 questions and add mixed temporal-contradictory and multi-timestamp categories.	Benchmark (future)
Y2	Threshold and parameter sensitivity studies (0.60-0.90; half-life 15-180 days; term ablation).	Planned evaluation
Y3	Scalability curve at 15 / 50 / 100 / 500 / 1,000 / 5,000 memories.	Planned evaluation
Y4	A learned rather than fixed weighting of the scoring terms, with the fixed version as baseline.	Future work

#	Change	Where
Y5	LLM-assisted extraction behind the high-precision rule-based trigger.	Future work
Y6	A user study on de-identified corpora, and on whether damping is preferred to removal by real users.	Future work

Stage 4 - Experimental Plan

Each experiment below is specified so that it can be run as written. Every one removes a [REQUIRES EXPERIMENT] marker from the revised paper. The baseline column names the comparison actually used, not an aspiration.

E1 - Component-wise ablation

Field	Specification
Objective	Attribute the measured effect to individual mechanisms rather than to the post-retrieval layer as a whole.
Research question	Which of the five mechanisms (temporal validity, contradiction damping, consolidation, forgetting, importance weighting) produces the MRR and stale@1 improvements?
Input / data	PersonalBrain-Bench as evaluated in the paper; same corpus, same questions, same candidate generation.
Baseline	Hybrid retrieval with the entire research layer disabled (row 1).
Proposed system	Cumulative rows, each adding exactly one mechanism: +temporal, +contradiction, +consolidation, +forgetting, +importance.
Variables	Independent: the five mechanism switches. Controlled: corpus, questions, k = 5, dense backend, corpus scope.
Metrics	Hit@5, MRR, stale@1, forgotten-leak rate, duplicate rate, latency.
Expected output	One row per configuration (6 rows), one table. Prediction to test: stale@1 falls on the temporal and contradiction rows; duplicate rate falls only on the consolidation row; leak rate only on the forgetting row.
Failure condition	If a mechanism's row shows no change on the metric it targets, the mechanism is not doing the work its description implies, and the paper must say so.

E2 - End-to-end generation evaluation

Field	Specification
Objective	Establish whether retrieval-level gains survive generation, which is the paper's actual claim about answers.
Research question	Do lower stale@1 and higher MRR produce fewer temporally wrong or contradicted answers in generated prose?
Input / data	The 11 benchmark questions, each run against all four pipelines.
Baseline	Same model, same prompt, same context length, with context supplied by each of the four retrievers.
Proposed system	Adaptive retrieval supplying the context.
Variables	Independent: retrieval pipeline. Controlled: model and version, temperature, prompt template, context budget, question order.
Metrics	Answer correctness, temporal correctness, contradiction resolution, evidence correctness, forgotten-content leakage in the generated text, abstention correctness. Two independent human raters, blind to the condition, with a stated disagreement rule (third rater or majority).
Expected output	Six metrics x four systems table, plus a per-question disagreement log.
Note	Automatic judges introduce their own error modes; if an LLM judge is used it must be reported with a human-agreement rate on a subset.

E3 - Conflict-detection accuracy

Field	Specification
Objective	Measure the detector's precision and recall, which are currently unknown; only its downstream effect has been observed.

Field	Specification
Research question	How accurate is three-class rule-based conflict detection on personal memory text?
Input / data	A manually annotated set of memory pairs: genuine conflicts by class (temporal, negation, value change) plus matched non-conflicts, including the negative cases the rules are most likely to confuse.
Baseline	The current rule-based detector.
Proposed system	Not applicable - this measures an existing component.
Variables	Independent: conflict class. Controlled: annotation guideline, annotator training, adjudication rule.
Metrics	Per-class precision, recall, F1; confusion matrix; false-positive examples quoted in the paper.
Expected output	3x3 confusion matrix plus a per-class metric table.
Known risk	Negation detection previously produced false positives by substring matching; recall is expected to be the weaker figure.

E4 - Threshold and parameter sensitivity

Field	Specification
Objective	Show that the consolidation threshold and the scoring constants are not arbitrary, and find where behaviour changes.
Research question	How sensitive are merge behaviour and retrieval quality to the similarity threshold, the recency half-life and the importance weights?
Input / data	A corpus enlarged enough to contain many duplicate and near-duplicate pairs (see E6); the current 15-memory corpus is too small for a threshold curve.
Baseline	Threshold 0.78, half-life 45 days, current weights.
Proposed system	Sweeps: threshold 0.60, 0.65, 0.70, 0.75, 0.78, 0.80, 0.85, 0.90; half-life 15, 30, 45, 90, 180 days; each weight zeroed in turn.
Variables	One parameter per sweep, all others held at the paper's values.
Metrics	Merge count, false merges (merged pairs a human judges unrelated), duplicate rate, Hit@5, MRR.
Expected output	One curve per sweep, plus a table of the value in use and the range over which results are stable.

E5 - Scalability

Field	Specification
Objective	Establish whether the per-run conflict and temporal lookups stay within an interactive budget as the store grows.
Research question	How do retrieval latency and offline maintenance cost grow with corpus size?
Input / data	Corpora of 15, 50, 100, 500, 1,000 and 5,000 memories, generated from the same templates with the same category distribution.
Baseline	Hybrid retrieval at each size.
Proposed system	Adaptive retrieval at each size.
Variables	Independent: corpus size. Controlled: hardware, database engine, question set, k.
Metrics	Retrieval latency per query (median and 95th percentile over repeated runs); offline time for scoring, conflict detection and consolidation; peak memory.
Expected output	Latency-versus-size curve for both pipelines, on named hardware.
Reporting rule	Latency is machine-dependent. Report hardware, give the curve, and do not claim cross-machine comparability.

E6 - Benchmark expansion and statistical protocol

Field	Specification
Objective	Make per-category claims statistically meaningful and remove the small-corpus limitation.
Research question	Do the observed differences hold beyond 11 questions and 15 memories?
Input / data	Proposed categories: factual, temporal, contradiction, multi-hop, duplicate, forgetting, abstention, and mixed temporal-contradictory cases with more than one supersession in the same predicate chain. Target at least 100 questions and several hundred memories, with the count per category stated.
Baseline	Three baselines as in the paper.
Proposed system	Adaptive pipeline.
Variables	Independent: pipeline. Controlled: corpus and question set, fixed across pipelines.
Metrics	All existing metrics, now reported as mean with standard deviation over multiple generated corpora, plus 95% bootstrap confidence intervals over resampled questions.
Statistical tests	Paired over the shared question set: Wilcoxon signed-rank for MRR-type statistics, McNemar's test for per-question hit and stale outcomes. State the number of questions and the significance level.
Expected output	One results table with CIs, one per-category table with counts and CIs, one statistical test table.

E7 - Removal completeness audit

Field	Specification
Objective	Replace the single-surface leakage check with a complete audit of every surface from which removed content could re-enter.
Research question	Which surfaces retain recoverable traces of a removed memory?
Input / data	A memory with a distinctive marker string, ingested, indexed, graphed, embedded and extracted into temporal facts; then removed.
Baseline	Row-deletion only: delete the memory row and query every surface.
Proposed system	The five-surface forgetting service.
Variables	Independent: removal strategy. Controlled: the memory and the query used to probe each surface.
Metrics	Per-surface recovery of the marker (binary), plus an end-to-end query test per retrieval mode.
Expected output	Surfaces x strategies matrix showing which combinations leave the content recoverable.
Note	This is also the experiment that justifies the phrasing “retrieval-level forgetting” rather than “machine unlearning”.

Stage 5 - Novelty Statement

The paragraph below is written to be defensible: each clause names what is claimed and distinguishes it from what is adopted from prior work. It contains no absolute priority claim, because establishing one would require a systematic review this work has not performed.

Novelty statement (one paragraph, for the introduction or the response letter)

This work addresses retrieval-augmented generation over *personal* knowledge stores, where the difficulty is not finding relevant evidence but determining which of several stored, individually faithful memories is currently true. The individual techniques it builds on are established and are not claimed as novel: dense, lexical and hybrid retrieval; graph-augmented retrieval; interval-based temporal triples; cosine near-duplicate clustering; and the broader problem of removing data from a system. What is proposed and evaluated here is (i) an operational treatment of current truth over a personal store - intervals that close facts rather than delete them, so that historical and current questions remain answerable from one store; (ii) *retrieval-time* enforcement of temporal validity, conflict status, duplication and explicit removal, rather than recording them as metadata, on the argument that retrieval time is where the failure occurs; (iii) a five-surface formulation of removal that treats derived indexes, vector embeddings, graph nodes and extracted facts as independent ways for removed content to return, which is a retrieval-layer property and is explicitly distinguished from parameter-level machine unlearning; (iv) an auditable importance score whose eight terms are persisted and which is used only as a light ordering signal, with the negative result that ranking by importance alone degrades retrieval reported rather than suppressed; and (v) PersonalBrain-Bench, a controlled corpus and question set containing superseded, contradictory, duplicated and deliberately forgotten memories, which makes these failures measurable in aggregate. The empirical contribution is a negative and a positive result together: on this corpus Hit@5 saturates for all four pipelines and therefore does not discriminate, while rank-sensitive and staleness measures do, with the proposed pipeline improving MRR and reducing stale@1 to zero in a reproducible, model-free configuration. The paper does not claim that its results generalise beyond the controlled corpus, and states the component-wise attribution, generation-level, detection-accuracy and scalability experiments that remain to be run.

Appendix A - Reference Audit

Every reference in the draft was checked against the published record. Entries marked “not located” could not be found under the given title, authors or venue, and should be removed unless the authors can supply the correct record. The revised paper's reference list (25 entries reduced to 21, renumbered in order of first citation) contains only locatable works that are actually cited.

Draft ref	Audit result	Action taken
[1] Lewis et al., RAG, NeurIPS 2020	Correct	Kept as [1]
[2] Gao et al., RAG survey, arXiv 2312.10997	Correct	Kept as [2]
[3] Edge et al., Graph RAG, arXiv 2404.16130	Correct	Kept as [6]
[4] Peng et al., Graph RAG survey, arXiv 2408.08921	Correct	Kept as [7]
[5] Wang et al., LLM autonomous agents survey, Frontiers Comput. Sci.	Correct	Kept as [10]
[6] Bruch, Nadelkoski, Mandal, vector stores, IEEE Big Data 2024	Verified in IEEE Xplore	Kept as [5]
[7] Hou et al., factual reliability, IEEE Access	Not located under this title/author combination	Removed
[8] Ji et al., RAG Certainty	The work exists and is on IEEE Xplore, but the year given in the draft (2026) is not supported by the record located. Confirm the venue and year from the paper's own IEEE Xplore entry before resubmitting.	Retained as [17]; year flagged for confirmation, not asserted
[9] “K. R. Z. et al.”, ReRag, IEEE Access 2025	Found, but author names are truncated to initials and the venue and year are wrong: Ko, Gürkan and Yarman Vural, UBMK (9th Int. Conf. Comput. Sci. Eng.), 2024, pp. 961-965	Authors, venue and year corrected; kept as [16]
[10] Asai et al., Self-RAG, ICLR 2024	Correct	Kept as [14]
[11] Yan et al., Corrective RAG, arXiv 2401.15884	Correct	Kept as [15]
[12] Prompt injection review, IEEE Access 2025	Not located as cited (the review found is dated later and the author list differs)	Removed - not cited in the argument anyway
[13] Zhu et al., LLMs concise review, IEEE Access 2025	Not located under this title; a different LLM review was found with different authors	Removed
[14] Brown et al., GPT-3, NeurIPS 2020	Correct, but not cited in the text	Kept as [20] and now cited where the LLM substrate is described
[15] Reimers & Gurevych, Sentence-BERT, EMNLP 2019	Correct, but not cited	Kept as [18] and cited in the retrieval section
[16] Karpukhin et al., DPR, EMNLP 2020	Correct	Kept as [3]
[17] Robertson & Zaragoza, BM25, FntIR 2009	Correct	Kept as [4]
[18] Johnson et al., FAISS, IEEE Trans. Big Data	Correct, but not cited	Kept as [19] and cited with the vector-store discussion
[19] Hogan et al., Knowledge Graphs, ACM CSUR	Correct	Kept as [8]

Draft ref	Audit result	Action taken
[20] Bourtole et al., Machine Unlearning, IEEE S&P 2021	Correct	Kept as [13] - now the anchor for the unlearning distinction
[21] Wu et al., complex reasoning survey, IEEE Trans. Artif. Intell. 2026	Not located as cited	Removed
[22] Zhang et al., personalised learning recommendation, IEEE Trans. Learning Technol. 2025	Not located under this title; a different paper with a similar title exists in a different journal	Removed
[23] "R. Wang et al.", Lifelong Learning of LLM-based Agents, TPAMI 2026	Real work, wrong authors: Zheng, Shi, Cai, Li, Zhang, Li, Yu and Ma, IEEE TPAMI 2026 (early access) / arXiv 2501.07278	Author list corrected; kept as [12] and now central to the agent-memory gap
[24] Li et al., Make LLMs Efficient, IEEE Access 2025	Not located as cited	Removed - not cited in the argument
[25] Radford et al., Whisper, ICML 2023	Correct, but not cited in the text	Kept as [21] and cited for the voice capture path

References added that the draft lacked

Four areas the argument depends on were unrepresented in the draft. These additions are all verified records and are cited where the corresponding claim is made:

- Temporal knowledge representation: Cai *et al.*, "Temporal Knowledge Graph Completion: A Survey", IJCAI 2023 (ref [9] in the revised paper). Without this, the temporal contribution reads as though interval validity were new.
- Agent memory mechanisms: Zhang *et al.*, "A Survey on the Memory Mechanism of Large Language Model based Agents", arXiv 2404.13501 (ref [11]). This is the closest prior work to the memory-persistence claim and must be cited to define the gap honestly.
- Lifelong learning for LLM agents: Zheng *et al.*, IEEE TPAMI 2026 (ref [12]) - the corrected version of the draft's [23].
- Machine unlearning: the draft's [20], now cited at the point where the distinction from retrieval-level forgetting is drawn.

Summary of Reviewer Verdict

The draft's central idea is sound and its framing is better than its execution. The blocking problems are not conceptual: two reported figures cannot be reproduced from the code, several references do not match the published record, and the contribution statement does not distinguish adopted techniques from proposed mechanisms. Each has been corrected in the accompanying revised paper.

The revised paper is, in my assessment, publishable at a conference as an *evaluation and architecture* paper with the small-corpus limitation stated prominently. It is not yet publishable as a systems result with general claims, because component-wise attribution, generation-level evaluation and detection accuracy are all unmeasured; the revised paper says so rather than implying otherwise. Completing E1 to E3 would raise it to a journal submission where the component attribution and the end-to-end result carry the argument.